Teaching a Neural Network the Glow of the Solar System

A geometrically constrained model of zodiacal light for NASA's SPHEREx mission

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A geometrically constrained neural network learns the zodiacal light from viewing geometry alone, reaching ~0.02% median error on months it never saw.

Background

- **SPHEREx:** NASA space telescope (launched March 2025) mapping the **entire sky in infrared** (0.75–5 μm)
- **Science:** how the universe began, how galaxies formed, where water and other ingredients for life are found

Zodiacal light

- Sunlight scattered by dust from comets and asteroids, plus the dust's heat glow (Fig. 1)
- Far brighter than the faint glow of distant galaxies SPHEREx aims to measure
- Changes with pointing and Earth's place in orbit → **must be modeled and removed**

The Kelsall model

- Standard dust model from NASA's COBE (Fig. 2); **ZodiPy** evaluates its prediction of the zodiacal-light contribution to what SPHEREx observes

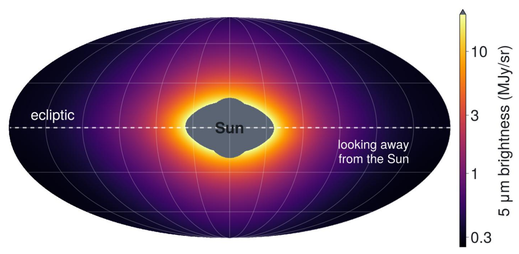


Figure 1. The whole sky at 5 μm from SPHEREx (ZodiPy): brightest toward the Sun and along the plane of the planets. Grey: too close to the Sun to observe.

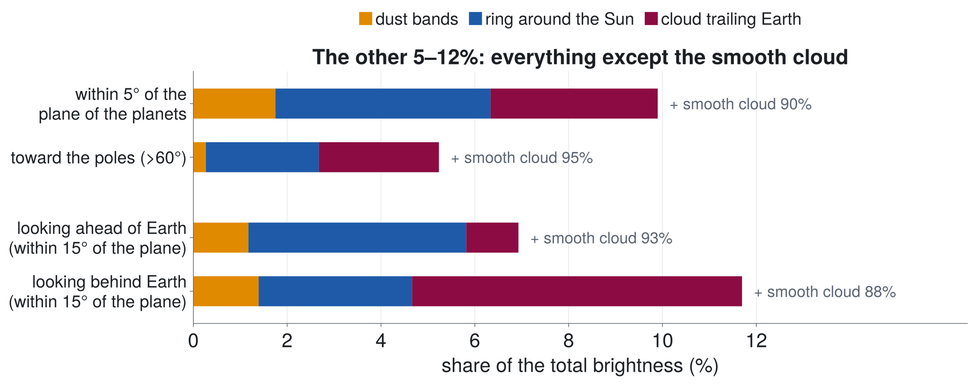


Figure 2. What makes up the zodiacal light SPHEREx sees (Kelsall model): a smooth cloud (88–95%) plus dust bands, a ring around the Sun, and a cloud trailing Earth, strongest when looking behind Earth.

The problem

- Zodiacal light must be removed from every SPHEREx image, but Kelsall's model uses a finite number of analytic components that don't fully describe the spatial and spectral structure in real data

Research objective

- A **geometrically constrained** neural network that learns zodiacal light from SPHEREx data and **outperforms the Kelsall model**. First step: ZodiPy's simulated sky

Methods

A geometrically constrained model

- No dates or sky coordinates: only **9 numbers describing the viewing geometry** (Fig. 3), dot products of the pointing vector n
- It must learn how brightness depends on geometry, as the physics does, not memorize when and where SPHEREx looked

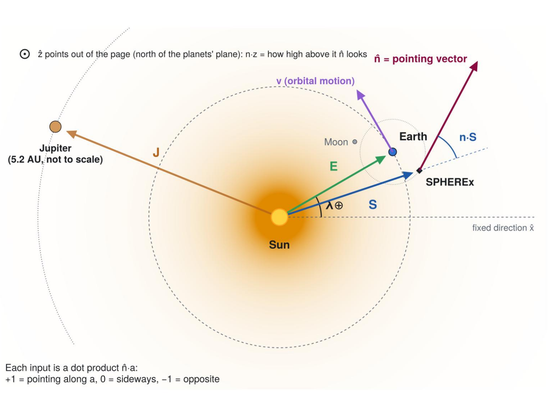


Figure 3. The geometry the network sees (top view, not to scale): n = SPHEREx's pointing vector; S, E = SPHEREx, Earth positions; v = orbital motion; λ⊕ = Earth's angle around its orbit. Jupiter and the Moon are tracked for real data: dust clouds leading and trailing Jupiter, and moonlight.

The 9 inputs

n·S, n·E	toward or away from the Sun (seen from SPHEREx and from Earth)
n·z	how far above or below the plane of the planets
n·vS, n·vE	ahead of or behind the orbital motion (SPHEREx, Earth)
n·x⊕, n·y⊕	Earth-following frame: x⊕ = Sun → Earth, y⊕ = Earth's motion
sin λ⊕, cos λ⊕	Earth's position in its orbit: the time of year

Training and testing

- 423,607 simulated pointings (140 days, 5 μm); five small networks (Flux.jl), averaged
- Tested on whole months and sky regions they never saw
- **Lesson:** adding the time of year (sin λ⊕, cos λ⊕) cut the errors ~40–65×

Results

0.018%
median error on months withheld from training

97.5%
of pointings within 0.1% of ZodiPy

≤0.03%
median error for every choice of withheld months

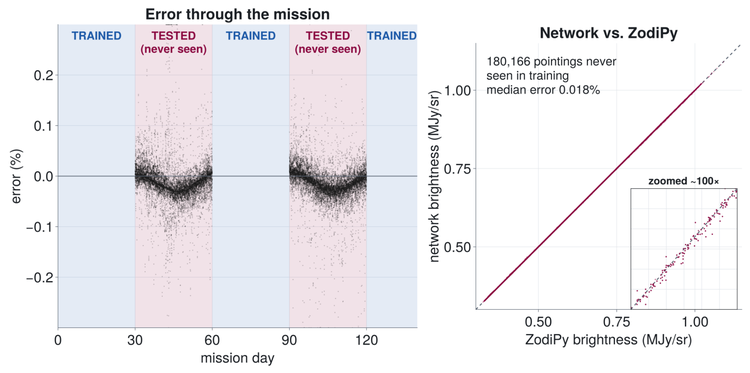


Figure 4. Network vs. ZodiPy on the two 30-day blocks never used in training (days 30–60, 90–120; 180,166 pointings). Left: error through the mission; right: every pointing falls on the 1:1 line.

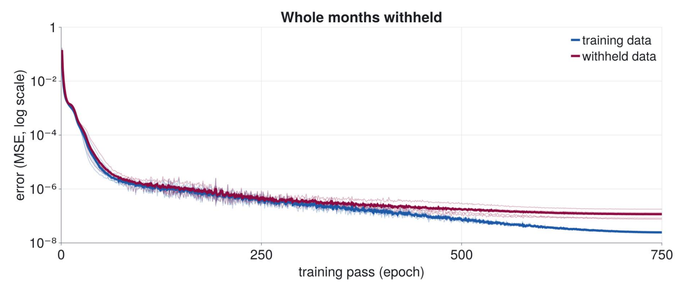


Figure 5. Error during training: 5 networks (thin), average (thick). The error on withheld months levels off and never rises: no overfitting.

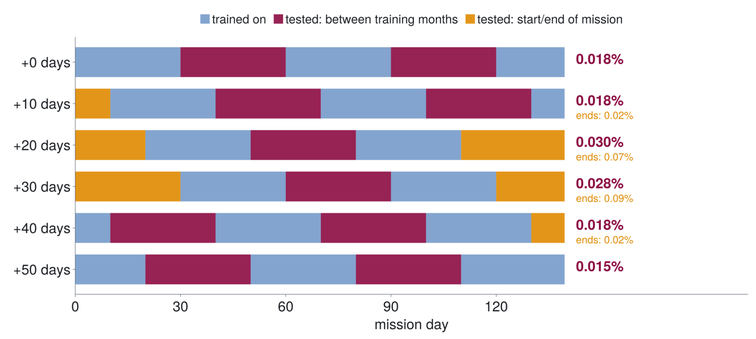


Figure 6. Six versions of the test, sliding the withheld months 10 days at a time: all reach 0.015–0.03% (red). Predicting past the start/end of the data (orange) is harder: fine for ~10 days, then errors grow.

- **Largest errors** along the plane of the planets, where the dust bands are (0.08%); **smallest** toward the poles (0.013%; Fig. 7)

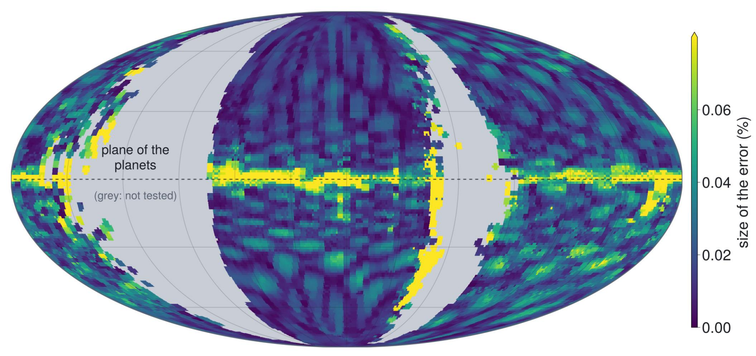


Figure 7. Error on pointings the network never saw, across the whole sky. Each pointing is drawn 3.5° wide, about SPHEREx's view. Grey: not tested.

Next Steps

✓ DONE Validate	Learn ZodiPy's simulated sky from geometry alone; test on unseen months and sky regions
IN PROGRESS Refine	Longer training; all SPHEREx wavelengths (a first 1.25–5 μm version: ~0.1% error)
NEXT Scale up	Move training to a computing cluster
NEXT Real data	Train on real SPHEREx images and compare with the Kelsall model

All results so far are preliminary and use simulated data.

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References

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